

【数据结构】链表

Tags:

Time Limit: 1 s Memory Limit: 128 MB
Submission: 211 AC: 31 Score: 98.53

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Codes

AI Code Tips

Description

链表的插入流程如下图所示：

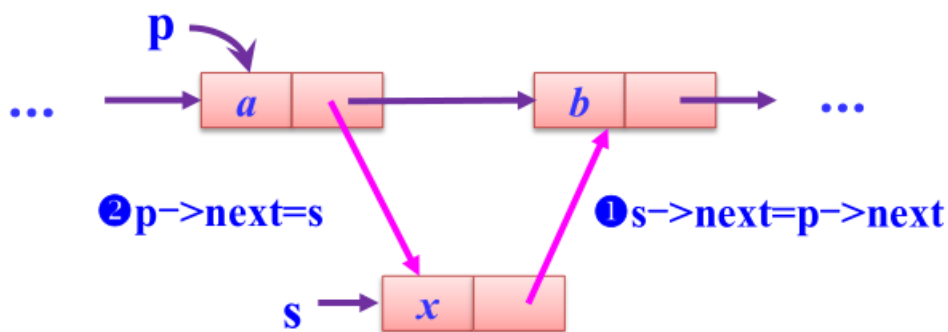
第一步：

新插入节点s的next指针指向p节点的next指针

第二步：

将p->next 指向s

单链表插入节点演示



插入操作语句描述如下：

① $s \rightarrow next = p \rightarrow next;$

② $p \rightarrow next = s;$

http://blog.csdn.net/weixin_38070406

完成以上两步之后，即完成了新节点的插入。

节点的删除过程与上图类似。

现在阅读以下程序，并补全

```
1 Status ListInsert(NHead *head, int pos, ElemType e);
2 Status ListDelete(NHead *head, int pos);
```

两个函数，使得程序能够完成新节点的插入或删除某个位置的元素

元素下标从0开始

```
1#include <stdio.h>
2#include <stdlib.h>
3#include <string.h>
4#include <ctype.h>
5#include <math.h>
6#define OK 1
7#define ERROR 0
8#define TRUE 1
9#define FALSE 0
10typedef int ElemType;
11typedef int Status;
12typedef struct Node
13{
14     ElemType data;
15     struct Node *next;
16}Node;
17typedef Node *NPointer;
18typedef struct NHead
19{
20     int length;
21     NPointer next;
22}NHead;
23Status InitList(NHead *head)
24{
25     head->length = 0;
26     head->next = NULL;
27     return OK;
28}
29Status ClearList(NHead *head)
30{
31     head->length = 0;
32     while (head->next)
33     {
34         NPointer tmp = head->next;
35         head->next = head->next->next;
36         free(tmp);
37     }
38     return OK;
39}
40Status ListIsEmpty(NHead *head)
41{
42     if (head->length)return FALSE;
43     else return TRUE;
44}
45int ListLength(NHead *head)
46{
47     return head->length;
48}
49Status GetElem(NHead *head, int pos, ElemType *e)
50{
51     NPointer now = head->next;
52     int cot = 0;
```

```

53 while (now->next < pos)
54 {
55     now = now->next;
56     cot++;
57 }
58 if (!now || cot >= pos) return ERROR;
59 *e = now->data;
60 return OK;
61}
62int LocateElem(NHead* head, ElemType e)
63{
64     int cot = 0;
65     NPointer now = head->next;
66     while (now)
67     {
68         if (now->data == e) /* 找到这样的数据元素 */
69             break;
70         now = now->next;
71         cot++;
72     }
73     return cot;
74}
75void ListVisit(ElemType e)
76{
77     printf("%d", e);
78}
79Status ListTraverse(NHead *head)
80{
81     NPointer now = head->next;
82     int cot = 0;
83     while (now)
84     {
85         if (cot) printf(" ");
86         ListVisit(now->data);
87         cot++;
88         now = now->next;
89     }
90     printf("\n");
91     return OK;
92}
93Status ListInsert(NHead *head, int pos, ElemType e);
94Status ListDelete(NHead *head, int pos);
95int main()
96{
97     NHead head;
98     InitList(&head);
99     int n;
100     scanf("%d", &n);
101     char ctl[10];
102     for (int i = 0; i < n; i++)
103     {
104         scanf("%s", ctl);
105         if (strcmp(ctl, "add") == 0)
106         {
107             int x, y;
108             scanf("%d%d", &x, &y);
109             ListInsert(&head, x, y);
110         }
111         else if (strcmp(ctl, "find") == 0)
112         {
113             int x;
114             scanf("%d", &x);
115             int pos = LocateElem(&head, x);
116             if (pos == ListLength(&head)) puts("No such element");
117             else printf("%d\n", pos);

```

```
118 }
119 else if (strcmp(ctl, "delete") == 0)
120 {
121     int x;
122     scanf("%d", &x);
123     if (ListDelete(&head, x))puts("OK");
124     else puts("ERROR");
125 }
126 else if (strcmp(ctl, "traverse") == 0)
127 {
128     ListTraverse(&head);
129 }
130 else if (strcmp(ctl, "clear") == 0)
131 {
132
133     ClearList(&head);
134 }
135 }
136 return 0;
137 }
```

你只需要提交两个函数即可

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Codes

HZNUOJ is based on HUSTOJ

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